

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	96.89%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.32%	Share of actual attacks successfully flagged
F1-Score	97.10%	Balanced measure combining Precision and Recall
Accuracy	98.16%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.45%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.68%	Actual attacks that went undetected
True Positive Rate (TPR)	97.32%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.55%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4575	FN 126
Actual: Normal	FP 147	TN 10017

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.98%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.68%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.32%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4575

2c. Hybrid Cascaded

Metric	Value	Description
Precision	96.89%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.32%	Share of actual attacks successfully flagged
F1-Score	97.10%	Balanced measure combining Precision and Recall
Accuracy	98.16%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.45%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.68%	Actual attacks that went undetected
True Positive Rate (TPR)	97.32%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.55%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4575	FN 126
Actual: Normal	FP 147	TN 10017

2d. Response Latency

Metric	Value	Description
Detection Time	567.9 ms	Average time taken to detect an attack
Mitigation Response Time	0.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.99%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	4.83%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.20%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	4	3	2026-09-26 18:45:27	2026-09-26 18:48:07
10.0.0.21	4	3	2026-09-26 18:45:31	2026-09-26 18:48:11
10.0.0.22	4	3	2026-09-26 18:45:25	2026-09-26 18:48:06
10.0.0.10	1	1	2026-09-26 18:44:42	2026-09-26 18:45:13
10.0.0.1	2	1	2026-09-26 18:44:29	2026-09-26 18:45:00
10.0.0.4	2	1	2026-09-26 18:44:27	2026-09-26 18:45:12
10.0.0.5	2	1	2026-09-26 18:44:26	2026-09-26 18:44:46
10.0.0.3	1	1	2026-09-26 18:45:01	2026-09-26 18:45:12

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 18:44:36	10.0.0.5	DDoS	Uncertain	58.8%	Low	Released
2026-09-26 18:44:38	10.0.0.4	DDoS	Uncertain	60.6%	Low	Released
2026-09-26 18:44:39	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 18:44:42	10.0.0.10	DDoS	UDP Flood	91.9%	High	Time Ban (30s)
2026-09-26 18:45:12	10.0.0.3	DDoS	Uncertain	60.6%	Low	Released
2026-09-26 18:45:13	10.0.0.10	DDoS	UDP Flood	91.9%	High	Released
2026-09-26 18:45:25	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:45:27	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:45:31	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:45:36	10.0.0.22	DDoS	UDP Flood	76.7%	High	Time Ban (30s)
2026-09-26 18:45:38	10.0.0.22	DDoS	UDP Flood	76.7%	High	Time Ban
2026-09-26 18:45:38	10.0.0.20	DDoS	UDP Flood	85.0%	High	Time Ban (30s)
2026-09-26 18:45:39	10.0.0.20	DDoS	UDP Flood	85.0%	High	Time Ban
2026-09-26 18:45:42	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (30s)
2026-09-26 18:45:42	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban
2026-09-26 18:46:37	10.0.0.22	DDoS	UDP Flood	80.4%	Critical	Time Ban (1m)
2026-09-26 18:46:39	10.0.0.20	DDoS	UDP Flood	87.2%	Critical	Time Ban (1m)
2026-09-26 18:46:42	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-26 18:48:07	10.0.0.20	DDoS	UDP Flood	82.6%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6463	77.7%	Critical	Phase 2	1	10s	2026-09-26 18:48:11	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	82.6%	Critical	Phase 2	1	10s	2026-09-26 18:48:07	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6435	75.8%	Critical	Phase 2	1	10s	2026-09-26 18:48:06	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6283	79.6%	High	Phase 2	3	1m 11s	2026-09-26 18:46:42	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6426	87.2%	Critical	Phase 2	3	1m 12s	2026-09-26 18:46:39	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6355	80.4%	Critical	Phase 2	3	1m 12s	2026-09-26 18:46:37	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.5959	79.6%	High	Phase 2	2	41s	2026-09-26 18:46:12	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6424	85.0%	High	Phase 2	2	42s	2026-09-26 18:46:09	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6402	80.4%	High	Phase 2	2	42s	2026-09-26 18:46:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6286	79.6%	High	Phase 2	1	11s	2026-09-26 18:45:42	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	85.0%	High	Phase 2	1	11s	2026-09-26 18:45:38	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	76.7%	High	Phase 2	1	11s	2026-09-26 18:45:36	Attack persisted - escalating to time ban
10.0.0.10	UDP Flood	0.6097	91.9%	High	Phase 2	1	31s	2026-09-26 18:45:13	Ban Expired
10.0.0.4	Uncertain	0.6125	60.6%	Low	Phase 1	1	11s	2026-09-26 18:45:12	Attack Stopped
10.0.0.3	Uncertain	0.6113	60.6%	Low	Phase 1	1	11s	2026-09-26 18:45:12	Attack Stopped
10.0.0.1	Uncertain	0.6111	58.9%	Low	Phase 1	1	11s	2026-09-26 18:45:00	Attack Stopped
10.0.0.5	Uncertain	0.6134	58.9%	Low	Phase 1	1	10s	2026-09-26 18:44:46	Attack Stopped
10.0.0.1	Uncertain	0.6105	58.9%	Low	Phase 1	1	10s	2026-09-26 18:44:39	Attack Stopped
10.0.0.4	Uncertain	0.5990	60.6%	Low	Phase 1	1	11s	2026-09-26 18:44:38	Attack Stopped
10.0.0.5	Uncertain	0.6118	58.8%	Low	Phase 1	1	10s	2026-09-26 18:44:36	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE